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Effect of Menaquinone-7 (MK-7) Supplementation on Anthropometric Measurements, Glycemic Indices, and Lipid Profiles: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

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ABSTRACT

Background: Menaquinone-7 (MK-7) is a type of vitamin K that has a longer half-life and stays in the body for a more extended period compared to other types of vitamin K. Recently, the effects of this vitamin on body weight, glycemic control, and lipid profiles have garnered much attention. **Aim of the review:** This systematic review and meta-analysis were performed to evaluate the effects of MK-7 on anthropometric measurements, glycemic indices, and lipid profiles.

Methods: A systematic search via appropriate keywords was conducted in electronic databases including PubMed, Scopus, Web of Science, and Google Scholar up to October 2023 to obtain relevant original articles. The quality of studies was evaluated using the Cochrane Collaboration tool. Six original articles met our criteria and were included in the analysis.

Results: Statistical analysis showed that MK-7 had a desirable effect on inulin (SMD= −0.56; 95 % CI: −0.77, −0.36; P = 0.000, I2 = 84 %, P = 0.000), HbA1c (SMD=-0.32; 95 % CI: −0.55, −0.09; P = 0.007, I2 = 86.8 %, P = 0.000), and homeostatic Model Assessment for Insulin Resistance (HOMA-IR) (SMD= −0.56; 95 % CI: −0.76, −0.35; P = 0.000, I2 = 84.3 %, P = 0.000). Additionally, subgroup analysis revealed negligible effects of MK-7 on total cholesterol (TC), insulin, HbA1c, and HOMA-IR in both genders of patients who received ≤ 90 mg MK-7 for less than 12 weeks. However, MK-7 didn't have any meaningful effect on other factors.

Conclusion: Based on the findings of the present systematic review and meta-analysis, MK-7 may have beneficial effects on glycemic control and TC, although further highly qualified original research is needed for a consistent conclusion.

Abbreviations: MK-7, Menaquinone-7; HOMA-IR, homeostatic Model Assessment for Insulin Resistance; TC, total cholesterol; K1vitamin, phylloquinone K2 vitamin, menaquinone; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analysis; RCTs, randomized controlled trials; SMD, standard mean difference; CI, confidence intervals; T2DM, type 2 diabetes mellitus; PCOS, polycystic Ovary Syndrome; FBG, fasting blood glucose; HOMA-B, homeostasis model assessment of β-cell dysfunction; LDL-C, low-density lipoprotein-cholesterol; HDL-C, high-density lipoprotein-cholesterol; TG, triglyceride; COC, Carboxylated osteocalcin; UcOC, Undercarboxylated osteocalcin; GPRC6A, G protein-coupled receptor family C group 6 member A; RDA, recommended dietary allowance; MCID, minimal clinically important difference.

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1. Introduction

The incidence of chronic illnesses, such as diabetes, cardiovascular disorders, and obesity, has been steadily increasing, posing significant challenges to both society and global healthcare systems [1]. The proper management of anthropometric measurements (e.g., BMI, weight), glycemic indices (e.g., HbA1c, HOMA-IR), and lipid profiles (e.g., cholesterol levels) plays a crucial role in maintaining general health and well-being [2]. These parameters serve as important markers of metabolic well-being and show a strong correlation with the likelihood of developing chronic ailments like diabetes, cardiovascular disorders, and obesity [3]. Consequently, there has been a growing interest in identifying effective interventions to improve these health indicators and reduce the burden of chronic diseases [4]. In recent times, there has been an upsurge of interest in the potential health benefits of micronutrients, such as vitamin K [5].

Vitamin K is a broad term encompassing multiple compounds, with phylloquinone (K1) and the K2 vitamins (menaquinones, MK) being of utmost significance [5]. Menaquinones (MK) are classified based on the length of their aliphatic side-chain, denoted by MK-n, where n signifies the number of isoprenyl residues present within the chain [5]. Vitamin K is found in various dietary forms, with phylloquinone (also known as vitamin K1) present in vegetables and MKs (referred to as vitamin K2) found in bacterially fermented foods [6]. In Japan, MK-4, one of the MK vitamins, is used as a pharmaceutical drug for the ethical management of osteoporosis [7]. MK-7, derived from the Japanese cuisine natto, specifically fermented soybeans, has demonstrated a longer half-life (3 days) compared to phylloquinone and MK-4, which have shorter half-lives (1–2 hours) [8]. Extensive meta-analysis research has been conducted on human subjects, consistently finding a connection between vitamin K and positive outcomes in various areas, including insulin resistance, abnormal lipid profiles, glucose tolerance, and even obesity [9,10]. Moreover, available evidence suggests that the hypoglycemic effects and the enhancement of anthropometric indices by MK-7 contribute to the improvement of blood lipid profiles [11–13].

To the best of our knowledge, there has not been an exhaustive examination of the impact of vitamin MK-7 on anthropometric measurements, glycemic indices, and lipid profiles through a systematic review and meta-analysis study. Therefore, it is of utmost importance to undertake such a systematic review and meta-analysis of existing studies to comprehensively assess the effects of MK-7 on these variables. Additionally, the outcomes derived from this systematic review and meta-analysis have the potential to offer valuable insights to healthcare professionals, policymakers, and researchers, thereby providing guidance for future research and interventions pertaining to the supplementation of vitamin K products. This systematic review and meta-analysis aim to comprehensively evaluate the effects of MK-7 on anthropometric measurements, glycemic indices, and lipid profiles, offering invaluable insights for healthcare professionals, policymakers, and researchers.

2. Method

This study conformed to the guidelines set forth in the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) [14], and adhered to the recommendations outlined in the Cochrane Handbook for Systematic Reviews of Interventions [15].

2.1. Registration

The systematic review and meta-analysis were duly registered in the

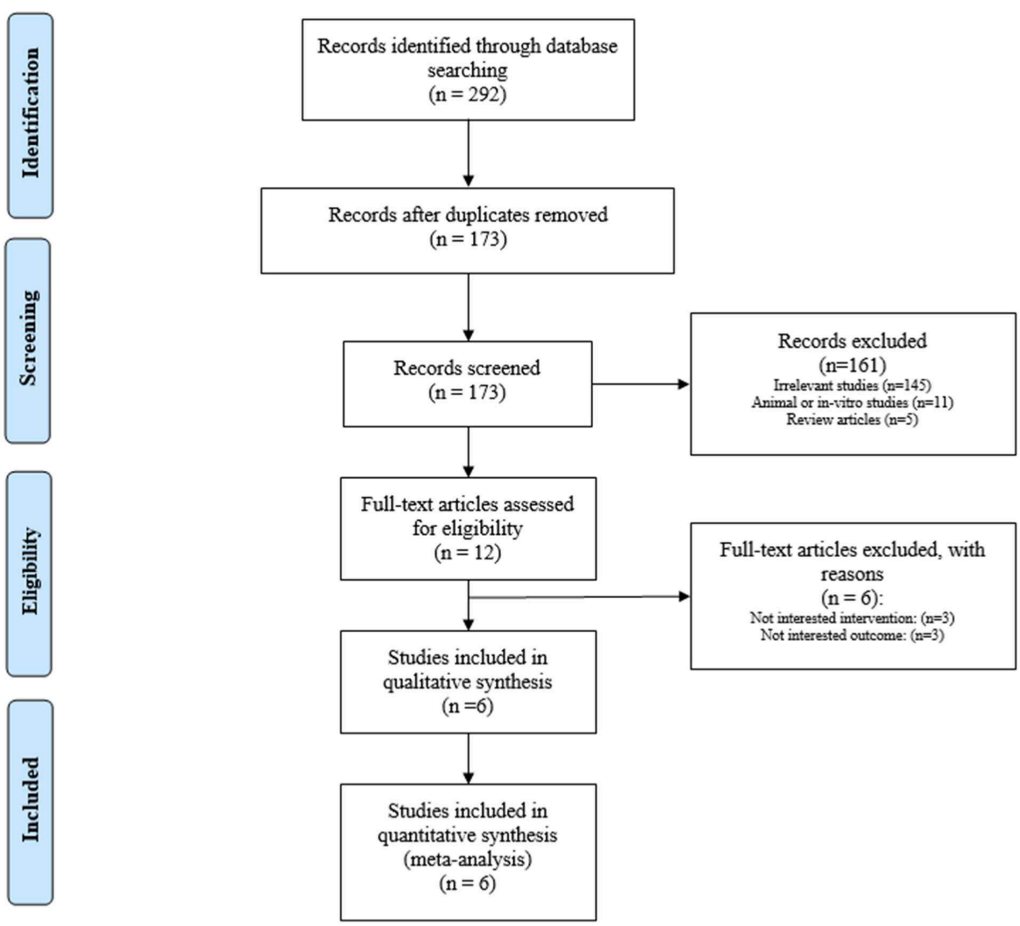

Fig. 1. Process of study selection.

Table 1
Characteristics of trials included in the meta-analysis.

Studies	Country	Study Design	Participant	Sample size and Sex	Sample size		Trial Duration (Week)	Mean ± SD or Median (IQR) Age		*Dose of supplement (mg/d)*		Type of supplement	
					IG	CG		IG	CG			IG	CG
Karamzad et al. 2020	Iran	RCT	Type 2 diabetes mellitus	M (31), F (14)	23	22	12	45.35 ± 6.25	46.95 ± 5.25	200 *μg*	Placebo	Menaquinone (MK−7)	Placebo
Tarkesh et al. 2020	Iran	RCT	Polycystic ovary syndrome	F (84)	42	42	8	28 (26–30)	27 (24–28)	90 *μg*	Placebo	Menaquinone (MK−7)	Placebo
Rahimi Sakak et al. 2021	Iran	RCT	Type 2 diabetes mellitus	—	32	31	12	58.5 ± 7.24	56.8 ± 7.27	360 μg	Placebo	Menaquinone (MK−7)	Placebo
Rønn et al. 2021	Denmark	RCT	Postmenopausal women with osteopenia	F (142)	71	71	48	68 ± 5	67 ± 4	375 μg	Placebo	Menaquinone (MK−7) with Ca and VitD	Placebo
Karamzad et al. 2022	Iran	RCT	Type 2 diabetes mellitus	M (31), F (14)	23	22	12	45.35 ± 6.25	46.95 ± 5.25	200 μg	Placebo	Menaquinone (MK−7)	Placebo
Zhang et al. 2023	China	RCT	Type 2 diabetes mellitus	M (29), F (31)	30	30	24	63.33 ± 1.33	62.97 ± 1.67	90 μg added in 100 g (1 cup) yogurt	Yogurt without MK−7	Yyogurt with menaquinone (MK−7)	yogurt without MK−7

Footprint: IG, intervention group; CG, control group; RCT, Randomised control trial

PROSPERO database, a well-established repository for systematic reviews. The registration number CRD42023431811 was obtained to ensure transparency and compliance with the review procedure.

2.2. Strategy and data sources

A thorough and methodical search was conducted to identify relevant studies from various databases, starting from the inception of the databases up to October 2023. The explored databases included PubMed, Scopus, and Web of Science. In addition to primary databases, alternative sources such as Google Scholar were explored to ensure a comprehensive search, as they may contain grey literature and studies not indexed in mainstream databases. The search strategy employed a combination of relevant keywords and search terms related to MK-7, measures of body size, indicators of glycemic levels, and lipid profiles. Logical operators (e.g., AND, OR) were used to ensure comprehensive results. The detailed exploration strategy is provided in Supplementary Table 1.

2.3. Inclusion and exclusion criteria and data extraction

Studies were selected for inclusion in the meta-analysis based on predetermined criteria, which considered the following factors: (a) only randomized controlled trials (RCTs) were included to ensure the presence of high-quality evidence, (b) studies involving human participants of any gender and health condition were considered, (c) studies that investigated the impact of MK-7 on anthropometric measurements, glycemic indices, and lipid profiles were incorporated. Studies were excluded from the analysis based on the following criteria: (a) studies conducted on non-human subjects or animal models, (b) articles not written in English, and (c) absence of relevant data or desired outcomes.

To extract pertinent information from the selected studies, a standardized data extraction form was employed. This process involved gathering details on study design, sample size, intervention duration, participant characteristics, specific details of MK-7 (dosage, frequency, and duration), outcome measures, and other relevant data required for the meta-analysis.

2.4. Data synthesis and quality assessment

The titles and abstracts of all records obtained from the search strategy were carefully reviewed by two reviewers (A.J and O.N), under the guidance of a senior reviewer (G.S), in accordance with the selection criteria. Detailed information regarding the studies, including study

Table 2
Quality of trials included in the meta-analysis.

Study, year	Random sequence generation	Allocation concealment	Blinding of participants & personnel	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other sources of bias	Overall quality
Karamzad, 2020	L	L	L	L	L	L	L	Low risk of bias
Karamzad, 2022	L	L	L	L	H	L	L	Fair
Rahimi Sakak, 2021	L	L	L	L	L	L	L	Low risk of bias
Rønn, 2021	L	L	L	L	L	L	L	Low risk of bias
Tarkesh, 2020	L	L	L	L	L	L	L	Low risk of bias
Zhang, 2023	L	L	L	L	H	L	L	Fair

Footprint: H, high risk of bias; L, low risk of bias; U, unclear risk of bias

Table 3
The results of categorical subgroup analysis for the effect of MK-7 on anthropometric measures, glycemic indices and lipid profiles.

Variables	No. of trial	SMD (95 % CI)	P-value	P for heterogeneity	I^2 (%)
BMI					
Total	4	0.00 (−0.21, 0.22)	0.979	0.118	48.9
Sex					
Female	2	−0.13 (−0.39, 0.13)	0.335	0.207	37.2
Both gender	2	0.27 (−0.12, 0.66)	0.169	0.227	31.4
Dose					
≤ 90	2	−0.01 (−0.34, 0.32)	0.968	0.015	83.0
> 90	2	0.00 (−0.29, 0.29)	1.0	1.00	0.00
Type of participants					
Diabetes	2	−0.13 (−0.39, 0.13)	0.335	0.207	37.2
Other	2	0.27 (−0.12, 0.66)	0.169	0.227	31.4
Duration of treatment	2				
≤ 12	2	−0.23 (−0.57, 0.12)	0.201	0.346	0.00
> 12	2	0.14 (−0.14, 0.42)	0.324	0.123	57.9
Weight					
Total	3	−0.02 (−0.26, 0.22)	0.888	0.786	0.000
WC					
Total	3	−0.01 (−0.29, 0.28)	0.970	0.837	0.000
Fat mass					
Total	3	−0.09 (−0.37, 0.20)	0.549	0.449	0.000
FBG					
Total	5	0.08 (−0.12, 0.27)	0.453	0.592	0.000
Sex					
Female	2	0.15 (−0.11, 0.41)	0.271	0.148	52.2
Both gender	3	−0.02 (−0.32, 0.28)	0.901	0.979	0.000
Dose					
≤ 90	2	0.23 (−0.1, 0.56)	0.167	0.238	28.2
> 90	3	−0.01 (−0.26, 0.24)	0.919	0.976	0.000
Type of participants					
Diabetes	3	−0.02 (−0.32, 0.28)	0.901	0.979	0.000
Other	2	0.15 (−0.11, 0.41)	0.271	0.148	52.2
Duration of treatment					
≤ 12	3	0.16 (−0.13, 0.44)	0.281	0.333	0.00
> 12	2	0.00 (−0.28, 0.28)	1.00	1.00	9.0
Insulin					
Total	5	−0.56 (−0.77, −0.36)	0.000	0.00	84.0
Sex					
Female	2	−0.31 (−0.58, −0.05)	0.021	0.00	92.7
Both gender	3	−0.92 (−1.25, −0.60)	0.000	0.239	30.1
Dose					
≤ 90	2	−1.13 (−1.49, −0.78)	0.00	0.408	0.00
> 90	3	−0.27 (−0.53, −0.02)	0.032	0.010	78.3
Type of participants					
Diabetes	3	−0.92 (−1.25, −0.6)	0.000	0.239	30.1
Other	2	−0.31 (−0.58, −0.05)	0.021	0.00	92.7
Duration of treatment					
≤ 12	3	−0.85 (−1.15, −0.56)	0.000	0.640	0.00
> 12	2	−0.30 (−0.58, −0.02)	0.039	0.000	94.1
HbA1c					
Total	4	−0.32 (−0.55, −0.09)	0.007	0.00	86.8
Duration of treatment					
≤ 12	2	−0.82 (−1.21, −0.42)	0.000	0.311	2.5
> 12	2	−0.06 (−0.35, 0.22)	0.650	0.00	92
HOMA-IR					
Total	5	−0.56 (−0.76, −0.35)	0.000	0.000	84.3
Sex					
Female	2	−0.28 (−0.54, −0.01)	0.040	0.001	90.8
Both gender	3	−0.96 (−1.29, −0.64)	0.000	0.121	52.7
Dose					
≤ 90	2	−1.11 (−1.46, −0.75)	0.00	0.124	57.7
> 90	3	−0.28 (−0.53, −0.02)	0.032	0.011	77.9
Type of participants					
Diabetes	3	−0.96 (−1.29, −0.64)	0.000	0.121	52.7
Other	2	−0.28 (−0.54, −0.01)	0.040	0.001	90.8
Duration of treatment					
≤ 12	3	−0.80 (−1.09,	0.000	0.875	0.00
> 12	2		0.025	0.000	95.0

(continued on next page)

Table 3 (*continued*)

Variables	No. of trial	SMD (95 % CI)	P-value	P for heterogeneity	I^2 (%)
		−0.51)			
		−0.33 (−0.61, −0.04)			
HOMA-B					
Total	3	−0.06 (−0.33, 0.21)	0.661	0.278	21.9
TC					
Total	5	−0.18 (−0.38, 0.03)	0.086	0.00	83.9
Sex					
Female	2	0.02 (−0.24, 0.28)	0.904	0.875	0.00
Both gender	3	−0.46 (−0.77, −0.14)	0.004	0.000	89.8
Dose					
≤ 90	2	−0.52 (−0.86, −0.18)	0.003	0.000	94.7
> 90	3	0.00 (−0.25, 0.25)	0.985	0.975	0.00
Type of participants					
Diabetes	3	−0.46 (−0.77, −0.14)	0.004	0.000	89.8
Other	2	0.02 (−0.24, 0.28)	0.904	0.875	0.00
LDL-C					
Total	4	−0.17 (−0.42, 0.08)	0.172	0.250	27
Dose					
≤ 90	2	−0.15 (−0.48, 0.18)	0.385	0.056	72.7
> 90	2	−0.21 (−0.59, 0.17)	0.279	0.534	0.000
HDL-C					
Total	4	−0.01 (−0.26, 0.24)	0.935	0.129	47
Dose					
≤ 90	2	0.12 (−0.21, 0.44)	0.489	0.116	59.6
> 90	2	−0.18 (−0.56, 0.20)	0.356	0.171	46.6
TG					
Total	4	−0.09 (−0.34, 0.16)	0.472	0.064	58.6
Dose					
≤ 90	2	−0.26 (−0.59, 0.07)	0.116	0.697	0.00
> 90	2	0.14 (−0.24, 0.52)	0.469	0.032	78.4

size, outcome measures, participants' baseline characteristics, intervention and comparison group details, and data collection and outcome reporting time points, was extracted from the full texts of the selected studies. This extraction process was carried out by the same two reviewers (P.N and M.P), and the extracted information was cross-validated for accuracy and consistency.

To assess the methodological quality of the included studies, the Cochrane Collaboration tool [15], as outlined by Higgins et al. [16], was utilized. This tool facilitated the evaluation of the risk of bias in the studies. Based on the criteria provided by the Cochrane Collaboration tool, the reviewers categorized the methodological quality of each study as either "low risk," "high risk," or "unclear risk." This assessment allowed for a comprehensive evaluation of the quality and potential biases present in the included studies.

2.5. Statistical analysis

The effect size was assessed using the standard mean difference (SMD), specifically Hedges' g, along with 95 % confidence intervals (CI). In cases where the studies did not provide standard deviations (SDs) for changes in outcomes from baseline, SDs were computed using the formula: SD = square root [(SD pre-treatment) ^{2} + (SD post-treatment) ^{2} - (2 R × SD pre-treatment × SD post-treatment)], assuming a correlation coefficient (R) of 0.5. For the meta-analysis, a random-effects model (DerSimonian and Laird) was used to pool the data based on the results of the heterogeneity test. The evaluation of heterogeneity involved the visual examination of forest plots, the $\chi 2$ test to assess the corresponding p-value, and the calculation of the I2 statistic. Considerable statistical heterogeneity was considered to exist if the p-value was less than 0.05 or if the I2 value exceeded 50 %. Sensitivity analyses were conducted to assess the robustness of the findings. Subgroup analyses was performed based on the duration of treatment, dosage of supplement, type of participants, and gender of subjects to evaluate the impact of these factors on the overall results. Furthermore, to assess publication bias, Begg's and Egger's tests were employed, accompanied by funnel plots for visual inspection of asymmetry. Stata software (StataCorp, College Station, TX, USA) version 15 was used to execute the statistical analyses.

2.6. Ethics and funding

Given that our investigation entails a methodical examination and meta-analysis of previously published research, no ethical acquiescence or informed consent was necessary. Additionally, the study protocol in Golestan University of Medical Sciences (IR GOUMS.REC.1402.218) was authorized by the regional ethical committee.

3. Result

3.1. Study characteristics

Overall 292 studies were obtained by systematic search in electronic databases, 119 articles were excluded due to duplication, and the title and abstract of 173 records were screened. 161 studies were not eligible due to the irrelevant studies (n = 145), animal or in-vitro studies (n = 11), and, review articles (n = 5). Therefore, we assessed the full text of 12 articles but 6 articles didn't meet our criteria due to not interested intervention (n = 3) and not interested outcome (n = 3), finally, 6 articles were entered into the final analysis. The flow chart of the selection study is presented in Fig. 1.

3.2. Study characteristics

Characteristics of the included studies are demonstrated in Table 1. All included studies in the present systematic review and meta-analysis have been published recently, from 2020 until 2023. Totally 439 (intervention group: 221, placebo group: 218) patients participated in all studies. The duration of treatment was between 8 and 48 weeks. The dosage of MK-7 used in the studies were as follows 90 μg [11,13], 200 μg [12,17], 360 μg [18] and 375 μg [19]. Out of all the included studies, four trials were conducted on type 2 diabetes mellitus (T2DM) patients [11, 12, 17, 18] and other researchers evaluated the effect of MK-7 on women with polycystic Ovary Syndrome (PCOS) [13] and osteopenia [19]. Quality assessment of studies revealed in Table 2, based on the Cochrane Collaboration tool four studies had low-risk bias [12, 13, 18, 19]and two studies were fair [11,17].

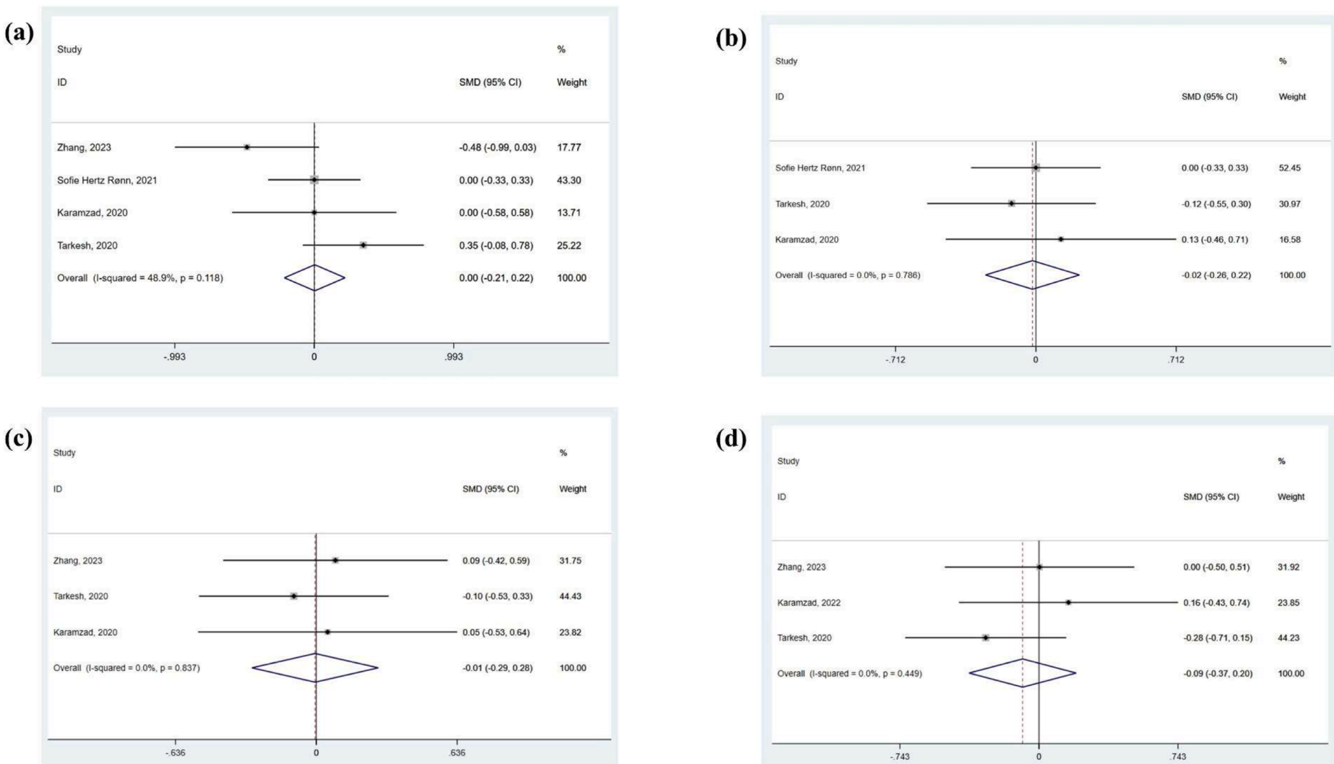

Fig. 2. Forest plot of the effects of MK-7 supplement on anthropometric measures (a: BMI, b: Weight, c: WC, d: Fat mass).

3.3. Effect of MK-7 supplementation on anthropometric measures

Fig. 2. demonstrate the effects of MK-7 anthropometric measures. Four studies evaluated the effect of MK-7 supplementation on BMI and the finding of statistical analysis indicated that MK-7 supplementation had no meaningful effect on BMI (SMD: 0.00, 95 %CI: −0.21, 0.22, p: 0.979) and heterogeneity between studies was not significant (I2:48.9, p: 0.118), Fig. 2. Also, 3 clinical trials reported the effect of MK-7 on the weight and pooled results didn't reveal any significant effect (SMD: −0.02, 95 %CI: −0.26, 0.22, p: 0.888) with negligible heterogeneity among the studies (I2: 00.0, p: 0.786). In addition, based on the results of 3 studies MK-7 didn't have any impressive effect on the WC (SMD: −0.01, 95 %CI: −0.29, 0.28, p: 0.970) and there was not any significant heterogeneity between studies (I2: 00.0, p: 0.837). Furthermore, 3 clinical trials evaluated the effect of MK-7 supplementation on fat mass and pooled effect size didn't show any significant effect (SMD: −0.09, 95 %CI: −0.37, 0.20, p: 0.549), and the heterogeneity between studies was insignificant (I2: 00.0, p: 0.449).

Sensitivity analysis demonstrated that the elimination of none of the studies had a significant on the pooled effects size of BMI, weight, WC, and fat mass. Also, statistical analysis didn't show any publication bias for BMI (Begg's $p = 0.999$ and Egger's $p = 0.829$), weight (Begg's $p = 0.602$ and Egger's $p = 0.977$), WC (Begg's $p = 0.602$ and Egger's $p = 0.363$), and fat mass (Begg's $p = 0.117$ and Egger's $p = 0.109$).

3.4. Effect of MK-7 supplementation on glycemic indices

Fig. 3. reveales the effects of MK-7 supplementation on glycemic control. Overall, five studies determined the effect of administration MK-7 on fasting blood glucose (FBG) and the analysis showed that mentioned micronutrient didn't have a significant effect on the FBG (SMD: 0.08, 95 %CI: −0.12, 0.28, p: 0.453), although there was not any heterogeneity between studies (I2: 00.0, p: 0.592). Based on the systematic search five studies evaluated the effect of MK-7 supplementation on the insulin serum level and pooled effect size showed a significant decrease (SMD: −0.56, 95 %CI: −0.77, −0.36, p: 0.000), but there was significant heterogeneity among studies (I2: 84.0, p: 0.000). Also, statistical analysis demonstrated that intake of MK-7 had a desirable effect on the HbA1c level (SMD: −0.32, 95 %CI: −0.55, −0.09, p: 0.007), although there was a high heterogeneity among entered studies (I2: 86.8, p: 0.000). In addition, the effect of MK-7 supplementation on homeostasis model assessment-estimated insulin resistance (HOMA-IR) was reported in 5 clinical trials and results of the statistical analysis demonstrated that it can significantly reduce the level of HOMA-IR (SMD: −0.56, 95 %CI: −0.76, −0.35, p: 0.000), although there was a high heterogeneity between studies (I2: 84.3, p: 0.000). Furthermore, 3 studies showed the effect of this vitamin on the homeostasis model assessment of β-cell dysfunction (HOMA-B). Statistical analysis revealed that MK-7 didn't have any significant effect on HOMA-B (SMD: −0.06, 95 %CI: −0.33, 0.21, p: 0.661) and between studies heterogeneity was trivial (I2: 21.9, p: 0.278).

Sensitivity analysis indicated that the pooled effect size of FBG, insulin, HbA1c, HOMA-IR, and HOMA-B does not change after excluding each of the studies. As well as, there was no publication bias for FBG (Begg's $p = 0.989$ and Egger's $p = 0.825$) and HOMA-B (Begg's $p = 0.602$ and Egger's $p = 0.284$), but based on the Egger's test there was publication for studies evaluate insulin (Begg's $p = 0.327$ and Egger's $p = 0.039$), HbA1c (Begg's $p = 0.042$ and Egger's $p = 0.007$), HOMA-IR (Begg's $p = 0.624$ and Egger's $p = 0.034$).

3.5. Effect of MK-7 supplementation on serum lipids

Fig. 4. indicates the effects of MK-7 on the lipid profile. Out of 6 included studies, five trials showed the effect of MK-7 on the level of total cholesterol (TC), but the effect of this vitamin on the level of low-density lipoprotein-cholesterol (LDL-C), high-density lipoprotein-cholesterol (HDL-C) and, triglyceride (TG) have been reported in four studies. Analysis according to the pooled effect sizes didn't indicate any remarkable effect on the level of TC (SMD: −0.18, 95 %CI: −0.38, 0.03, p: 0.086) while there was a high heterogeneity among studies (I2: 83.9,

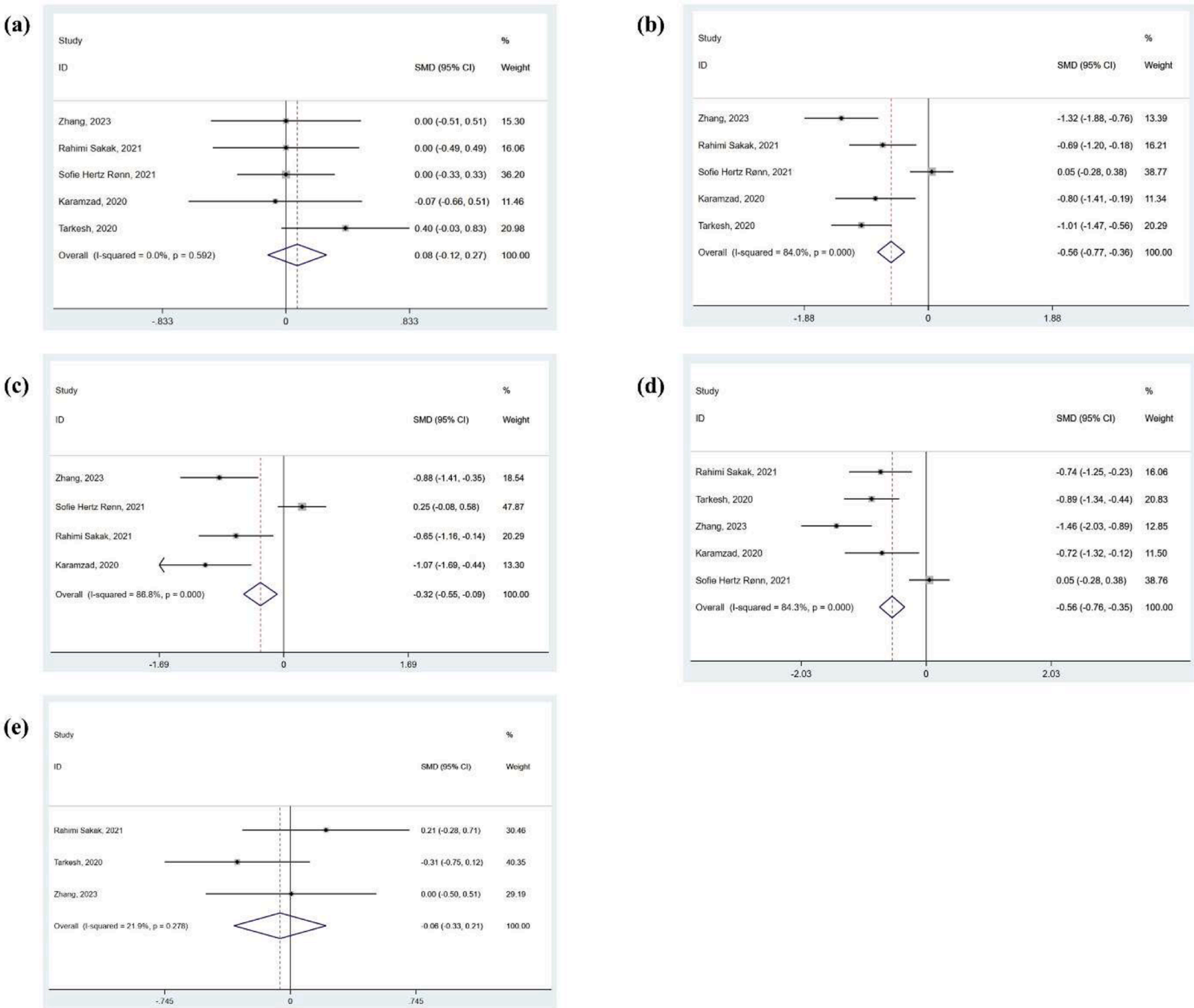

Fig. 3. Forest plot of the effects of MK-7 supplement on glycemic indices (a: FBG, b: Insulin, c: HbA1c, d: HOMA-IR, e: HOMA-B).

p: 0.000), LDL-C (SMD: −0.17, 95 %CI: −0.42, 0.08, p: 0.172), although heterogeneity between studies wasn't significant (I2: 27, p: 0.250), HDL-C (SMD: −0.01, 95 %CI: −0.26, 0.24, p: 0.935), however there wasn't a high heterogeneity between studies (I2: 47, p: 0.129) and, TG (SMD: −0.09, 95 %CI: −0.34, 0.16, p: 0.472) with a meaningful heterogeneity between studies (I2: 58.6, p: 0.064).

Sensitivity analysis showed that the exclusion of any of the studies does not have an impressive effect on the overall finding of TC, LDL-C, HDL-C, and TG. In addition, there was no evidence of publication bias for studies assessment the effect of this micronutrient on the level of TC (Begg's p: 0.05 and Egger's p:0.386), LDL-C (Begg's p:0.174 and Egger's p: 0.180), HDL-C (Begg's p: 0.463 and Egger's p:0.527), and TG (Begg's p: 0.497 and Egger's p: 0.320).

3.6. Subgroup analysis

MK-7 demonstrated a more significant effect on insulin and HOMA-IR when diabetes patients received ≤ 90 mg for less than 12 weeks, as well as on HbA1c for ≤ 12 weeks. Furthermore, this micronutrient had a beneficial effect on the TC of diabetes patients receiving ≤ 90 mg MK-7.

4. Discussion

The present systematic review and meta-analysis revealed for the first time that MK-7 supplementation has no significant effect on anthropometric indices and lipid profile. However, this supplement can improve insulin resistance and plasma insulin levels especially in diabetics and in doses less than 90 mg per day for ≤ 12 weeks as well as HbA1c for ≤ 12 weeks. Significant results were observed in some studied subgroups of TC. However, due to the high heterogeneity as a result of the difference in methodology and study population, these results should be interpreted with caution and more studies are needed in this field.

Vitamin K has been found to act as a key element in the gamma-carboxylation of osteocalcin [20]. Osteocalcin is a protein primarily produced by osteoblasts, the cells responsible for bone formation. It plays a crucial role in bone metabolism and mineralization. Osteocalcin is synthesized as a precursor molecule and undergoes a process called carboxylation, which involves the addition of carboxyl groups to glutamic acid residues. This carboxylation process is necessary for the full activation of osteocalcin. Carboxylated osteocalcin (cOC) is the fully activated form of the protein and is involved in regulating bone mineralization and turnover. It binds to hydroxyapatite, a mineral

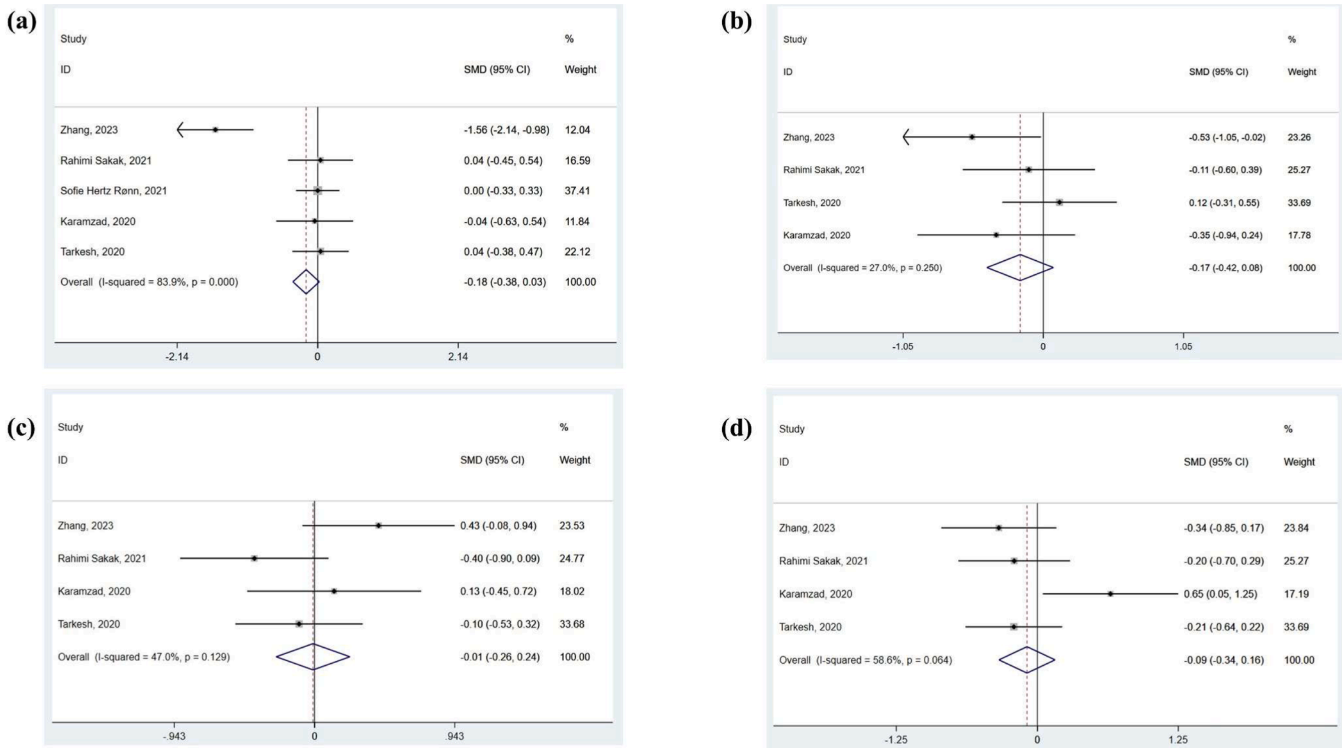

Fig. 4. Forest plot of the effects of MK-7 supplement on serum lipids (a: TC, b: LDL-C, c: HDL-C, d: TG).

component of bone, and promotes the deposition of calcium and phosphate ions, which are essential for bone strength and structure [21]. In addition to its role in bone health, osteocalcin has been found to have effects on other tissues and organs, including the pancreas, adipose tissue, and muscles. It is believed to act as a hormone and a signaling molecule, influencing various physiological processes [22]. One of the key roles of osteocalcin is its involvement in glucose metabolism and insulin sensitivity. Studies have shown that cOC can interact with pancreatic beta cells, stimulating insulin secretion [23,24]. The mechanisms by which osteocalcin influences glucose metabolism and insulin sensitivity are not fully understood. However, it is believed that osteocalcin may activate signaling pathways involved in insulin secretion and glucose uptake through increasing adiponectin secretion in adipose tissue as well as glucagon-like peptide-1 secretion in small intestine [24]. Undercarboxylated osteocalcin (ucOC) is a form of the protein osteocalcin that has not undergone a process called carboxylation, which is necessary for its full activation. ucOC has not affinity to bind calcium, therefore, it is secreted into the blood stream, binds to G protein-coupled receptor family C group 6 member A (GPRC6A), and has been found to have effects on other tissues and organs, including the pancreas and adipose tissue. It is believed to act as a hormone and a signaling molecule, influencing various physiological processes [25–28]. One of the key roles of ucOC is its involvement in glucose metabolism and insulin sensitivity. Studies have shown that ucOC can interact with pancreatic beta cells, which are responsible for producing and releasing insulin. This interaction stimulates insulin secretion, leading to increased uptake of glucose by cells [29]. Various human studies have determined that cOC and ucOC levels are low in diabetic patients and there is an inverse relationship between the levels of these compounds and blood sugar and insulin [30–32]. However, the ucOC/cOC ratio is a stronger indicator for the prognosis of diabetes. Villafán-Bernal et al. reported that a ucOC/cOC ratio less than 0.31 is associated with poor metabolic control of diabetes [33]. Vitamin K supplementation could reduce ucOC and increase cOC [19, 34, 35]. However, Aguayo-Ruiz et al. showed that vitamin K supplementation led to the mean value of ucOC/cOC ratio > 1, which does not approximate the risk value [36]. Therefore, positive effect of vitamin K on insulin and insulin resistance index can be mediated by its improving effect on cOC level. The greater effect at lower doses ($<$90 mcg/day) of MK-7 by subgroup analysis could be due to the possible effect of higher doses on this ratio leading to a greater reduction of ucOC. Therefore, it is recommended that vitamin K supplementation must be limited to the recommended dietary allowance (RDA) (120 mcg/day for males and 90 mcg/day for females) [37] to have a more improving effect on insulin resistance. However, the Food and Nutrition Board reported that "no adverse effects associated with vitamin K consumption from food or supplements have been reported in humans or animals" [37].

Clinically, we found that MK-7 did not have an important effect on insulin levels (our effect size for insulin is -0.56 μIU/mL which equals to 3.36 pmol/L based on the correct unit conversion for human insulin of 1 μIU/mL $= 6.00$ pmol/L reported by Knopp et al. [38]) according minimal clinically important difference (MCID) reported by previous meta-analysis (-5 pmol/L) [39], which led to a non-significant effect on fasting blood sugar levels, lipid profile and anthropometric indices, which are all affected by insulin resistance and insulin function. Moreover, MCID was determined 0.5 % or 5 mmol/mol for HbA1c [40]. However, we found that MK-7 effect size on HbA1c was less than recommended MCID. Therefore, it is recommended that MK-7 should be taken at the RDA dose only as an adjunctive approach beside main treatments to improve insulin resistance and HbA1c.

Although based on the present study MK-7 administration had beneficial effects on anthropometric measures, glycemic control, and lipid profile, it should be noted that there are other ingredients such as antioxidants and fibers especially soluble fibers which may had desirable effects on these variables. According to the Rahmani et al. study it was exhibited that cereal beta-glucan as a soluble fiber significantly decreased the body weight and BMI. Possible mechanism of soluble fiber associated with delaying upper gastrointestinal emptying and stimulation cholecystokinin (CCK) secretion [41]. It seems that, for a better-consolidated decision possible interactions between various nutrients should be considered. Given that, the present study is a systematic review and meta-analysis we could not adjust the effects of other confounders.

We had various limitations in conducting this meta-analysis. First, the number of studies in this field was small, and as a result, additional subgroup analyses and detailed analysis of publication bias with funnel plots were not performed. Also, due to the small number of studies, it was not possible to exclude studies that led to high heterogeneity. Consequently, some interpretations should be made with caution. In addition, meta-regression analysis was not performed to find confounding variables due to the small number of studies. Second, sample size in most of the included studies is relatively small. Therefore, in most cases, we cannot achieve sufficient power to find potentially significant effects. Consequently, additional studies with larger sample sizes may contradict our findings. It is recommended that more studies in the future measure the effect of vitamin K supplementation on anthropometric indices in people with high BMI and lipid profile in people with dyslipidemia to prove the lack of effect of this supplement on these indices. Also, studies in this field should also measure the ratio of ucOC/cOC in order to reach a more complete conclusion.

5. Conclusion

Based on the outcomes of the current systematic review and meta-analysis MK-7 probably had a desirable effect on insulin, HOMA-IR, HbA1c, and TC. However, the analysis didn't indicate a significant effect on weight, BMI, WC, fat mass, FBG, HOMA-B, LDL-C, HDL-C, and TG. A definitive conclusion about the effects of MK-7 on anthropometric variables and metabolic indices requires further research.

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CRediT authorship contribution statement

Pasand Mohammadjavad: Writing – review & editing, Investigation, Data curation. **Noura Pardis:** Writing – review & editing, Investigation, Data curation. **Sohrab Golbon:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis. **Nikpayam Omid:** Writing – review & editing, Validation, Project administration, Methodology, Formal analysis, Conceptualization. **Jafari Ali:** Writing – original draft, Investigation, Data curation. **Faghfouri Amir Hossein:** Writing – original draft, Investigation, Data curation.

Declaration of Competing Interest

None

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Not applicable

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.prostaglandins.2025.106970.

Data availability

Data will be made available on request.

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